

Section 7.2: Data Handling Applications

Computers (and computer applications) are often used to perform data handling tasks.

These tasks include:

- 1 Surveys 2 Address Lists 3 Tuck Shop Records
- 4 Record Keeping 5 School Reports 6 Library Data

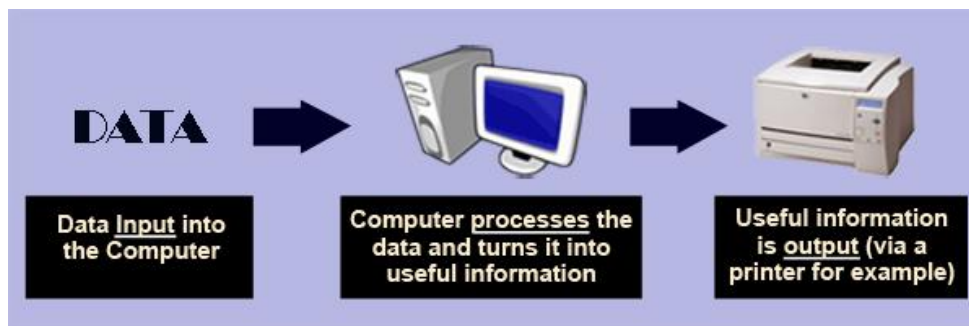
Key Concepts of this section:

- # Understand what '**data handling**' is .
- # Know why computers are **more effective at handling data** than paper-based methods.
- # Be able to discuss the **advantages** of using a computer to handle data in a **range of different scenarios**.

Data Handling

What is Data Handling?

- # Data Handling is basically where data is **input and stored** on a computer. The data is then **processed in some way** to turn it into useful information. This information can then be **output and used**.



Key Words:

Input, Processing, Output, Searching, Backup, Editing, Security, Sorting



Computer systems of all kinds can be used to handle data

Why use a computer to handle data?

Computers are much **better at handling data** than paper-based methods (like files of paper) for many reasons including:

- **Searching** - Computers can find stored data very quickly
- **Editing** - Computer data can be edited (changed) very easily
- **Backed up** - Copies of data can be made very easily
- **Security** - Passwords can be applied to computers
- **Sorting** - Data can be organised in different orders quickly
- **Transported** - Computer data can be moved via flash memory
- **Output** - Data can be printed out easily

NOTE!

Computers can also hold **vast amounts of data** in **very little physical space**. For example - A database on a standard home PC can store hundreds of filing cabinets worth of data!!



Paper-Based records can take up **a lot** of physical space.

Examples of Data that could be Stored on a Computer



Key Words:

Surveys, Address Lists, Shop Records, Club Records, School Reports, Library Records

The following examples show you how data can be stored and handled on a computer by using various every-day scenarios:

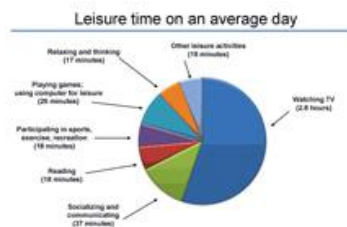
1 Survey Results

- # Surveys are often used by businesses / organisations to **find out what people want or do**.
- # Surveys are often carried out on paper and are filled in by shading circles or ticking boxes.
- # The paper-survey's data can then be scanned into a computer by using an **Optical Mark**

Reader (OMR). This eliminates typing the data in by hand!!

When the data is inside the computer it can be used in a number of ways:

- **Sorted** - e.g. ordered by date
- **Searched** - specific data can be found quickly
- **Analysed** - via graphs or charts
- **Results** - the findings of the survey can be published as a report



Gathering data using a survey

The benefits of using a computer to handle survey data are as follows:

- **Faster** - results of the survey
- **Less Errors** - input errors are minimal
- **Fast Results Analysis** - data analysed quicker
- **Time/Money Savings** - fewer people are needed to carry out the survey

NOTE!

Surveys are also carried out **via web sites**. Online surveys do not need to be entered into a computer via an OMR as the **data is sent automatically to a database**.

Click [here](#) for an example of an online survey.



Optical Mark Readers can **quickly scan the results** of a survey **into a computer**

2 Address Lists

Address lists are used to store **details about friends, family or business contacts**.

The details that we might store include:

- **Names**
- **Addresses**
- **Telephone Numbers (home, work and mobile)**
- **Email Address**
- **etc.....**

Before we had computers, this type of information was often stored in an **address book**



like the one in the image to the right. This sometimes caused problems like:

- **Easy to lose** - paper address books are quite small and easy to misplace
- **Hard to update** - you had to cross old entries out and replace
- **Hard to search** - if you have a lot of contacts

Modern address lists are **stored on computers** or **mobile phones** and this offers a number of advantages including:

Advantages of Electronic Address Lists

- Groups** - ➔ Family data can be **placed in one group**, friends in another group. This makes **finding contacts quicker**.
- Easy Editing** - ➔ If a contact changes (e.g. a new phone number) you can **easily edit the contact**. On a paper address book you have to cross out the old entry.
- Easy Searching** - ➔ Computerised address books always have **search facilities** which are much **quicker** than looking through pages of a paper book.
- Backups** - ➔ Computerised address books can be **easily copied** in case anything happens to the original.

3 Tuck Shop Records

Tuck shops sell things like **sweets, drinks, chocolate** and **crisps** to children.

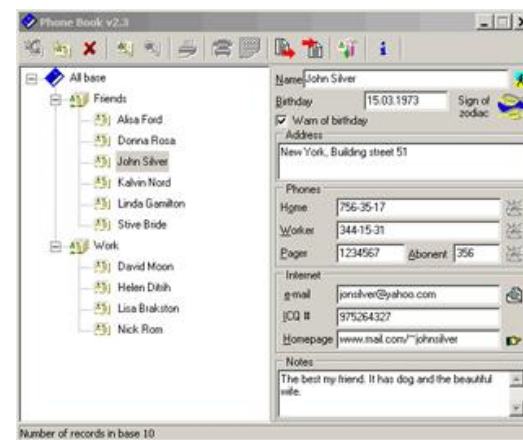
Like any type of shop, tuck shops need to keep a track of records such as:

- **Income** - daily takings
- **Expenditure** - costs to the business (buying products to sell, shop rent etc)
- **Stock Levels** - how much of each product are left
- **etc.....**

Before computers, these records were kept on paper in something called an '**Account's Ledger**'. As with any paper-based data handling method, accounts ledgers have certain drawbacks:

- **Easy to lose** - physical books can be misplaced
- **Hard to update** - old records would be crossed out and replaced (messy)
- **Difficult to backup** - any copies had to be made by photocopier

Paper address lists are **hard to update** and **easy to lose**



Electronic address lists can easily **group contacts**, **search contacts** and **edit contacts**



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- **Can be damaged - fire, water etc**
- **Entry errors - easy to miscalculate or write in the wrong information**

These days, spreadsheets are used to store shop account records. Spreadsheets offer a number of advantages when used to keep shop records:

Advantages of keeping shop records on a spreadsheet

Calculations / Formulae -



Calculations can be made **automatically** to give us **accurate totals and averages** etc. They can also **predict** how much of a profit or loss the shop will make.

Charts / Graphs



Charts can be quickly made to show us **changes** in things like **income, expenditure and profit/loss**.

Easy Editing -



Figures can be **changed easily** by simply replacing the old data. On a paper accounts ledger you have to cross out the old entry.

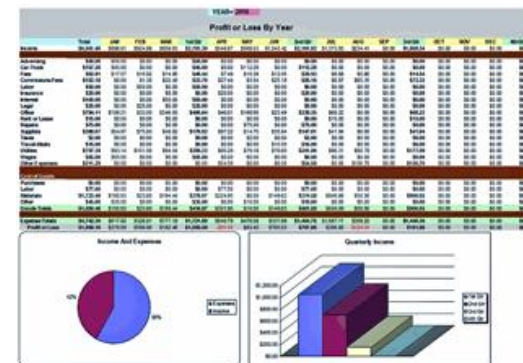
Backups -



Spreadsheets can **easily be copied** in case anything happens to the original.



Paper-based accounts records have **drawbacks**



Spreadsheets can **perform calculations** and are **easy to update**

4 Clubs and Society Records

Clubs and societies often have **many members**. Records need to be kept about these members for details such as:

- **Membership number**
- **Names**
- **Addresses**
- **Telephone number**
- **Email address**
- **Expiry date of membership**
- **Membership fee's paid (Y/N)**

Computer databases (e.g. Microsoft Access) are often used to store these details and this offers a number of advantages compared with storing the data on paper:



A club membership card

Advantages of storing membership details on a Database

- Easy Searching -** ➔ Member data can be searched very quickly. This is very useful if the club has hundreds or thousands of members.
- Easy Editing -** ➔ Member data can be changed easily. Old records (like address) are just deleted and replaced with the new records.
- Backups -** ➔ Databases can be copied very easily.
- Security -** ➔ Databases can be password protected. This helps stop unauthorised viewing of member's personal details (like address and phone numbers etc).
- Moving the Database -** ➔ If you need to move the database to a different location you just copy it to a portable storage device (e.g. flash memory) and take it with you.
- Mail Merge -** ➔ Databases can be linked with Word Documents to make it easy to create letters to be sent to all of the club's members.



Computerised databases are easy to **search**, **update**, **backup** and can be used to **perform mail merges**

NOTE!

This video ([link here](#)) shows you how **mail merge** can be used to **quickly create letters** to be sent to all of the members stored in your database.

5 School Reports

- # Schools keep records about **how well students are performing in their classes**. These records are shared with both the student and their parents each term.

The details that would be recorded include:

- **Student ID**
- **Student name**
- **Tutor group**
- **Subject (like ICT, English etc)**
- **Current grade**
- **Effort level**
- **Behavior level**
- **Effort level**
- **Homework level**
- **Teacher comments**

- # Before we had computers, student's reports had to be **written by hand**. This created problems such as:



Hand written reports were a **hugely frustrating** and **time consuming** task for teachers

- **Time** - hugely time consuming
- **Spelling** - with no spell check spelling errors were common
- **Errors** - any mistakes meant re-writing the report

Modern school reports are entered and stored in a database (like Microsoft Access). This gives many advantages when compared to writing them by hand:

Advantages of using a Database to create school reports

- Pre-Written Comments** - ➔ Pre-Written comments can be stored in the database and then **selected from a list**. The teacher can just select the comment that best matches the student instead of writing one from scratch. This is much **quicker and easier**.
- Easy Editing** - ➔ If a mistake is made you can simply **delete it and re-type**.
- Spell Check** - ➔ Database software include **spell checkers** which help make sure the reports are **grammatically correct**.
- Security** - ➔ Databases can be **password protected**. This means that personal student data is safe from **unauthorised viewing**.
- Reports** - ➔ Data stored in the database can be **displayed on reports**. Reports can be **re-designed** to look very **professional** and include **school logo's** etc.

Academic Report 2012-2013										
IPC	2A									
	Well Below Exp	Below Exp	At Exp	Above Exp	Well Above Exp	Well Below Exp	Below Exp	At Exp	Above Exp	Well Above Exp
Graduation 88 %						Graduation				
Teacher Comments: Roshni is working well above expectations in IPC. She has enjoyed learning about the five food groups and is able to discuss the topic confidently with specific vocabulary. She is also able to identify some countries on a world map and enjoyed using ICT to investigate different types of maps.						Teacher Comments:				
Mathematics	2B									
	Well Below Exp	Below Exp	At Exp	Above Exp	Well Above Exp	Well Below Exp	Below Exp	At Exp	Above Exp	Well Above Exp
Graduation 82 %						Graduation				
Teacher Comments: Roshni is working well above expectations in Maths. She is able to tell the time to the half hour and quarter hour and is able to solve time word problems. She enjoyed our measurement unit and can estimate and measure different lengths using formal and informal units.						Teacher Comments:				

Databases allow us to produce **professional looking student reports** at a **fraction of the effort** of writing them by hand

6 School Library Records

Libraries need to keep track of the following .

- **Which books are currently on loan**
- **ID number of the borrower**
- **Name of the borrower**
- **Address of the borrower**
- **Date the book was borrowed**
- **Date the book is due to be returned**

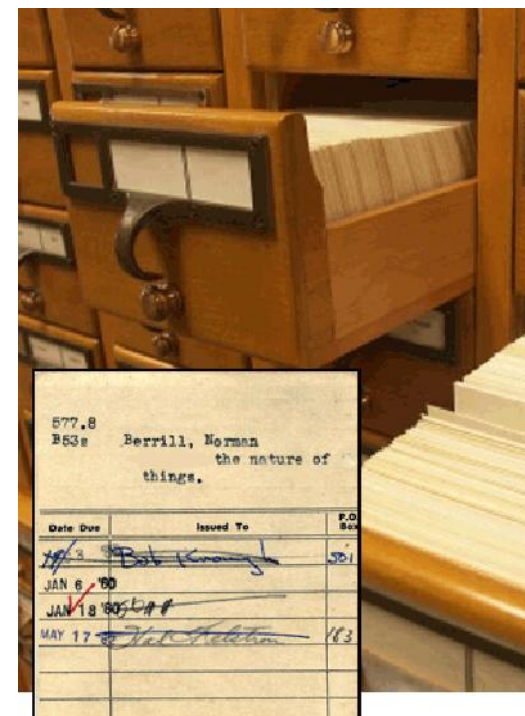
Before computers, book details were **recorded on cards** which were stored in organised filing drawers. This paper-based system posed problems including:

- **Errors** - manually checking the cards meant that sometimes overdue books were missed
- **Mix-ups** - cards could get mixed up or placed in the wrong drawer
- **Time** - it took lots of time to find the details of each book
- **Space** - large libraries might have thousands of paper-based book records. These took up a lot of room to store.

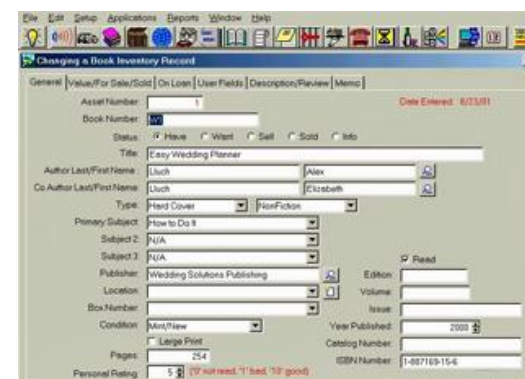
These days, libraries use databases to keep track of their books. Storing book and borrower records on a database give certain advantages including:

Advantages of storing library records in a database

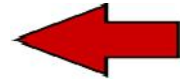
- Speed** - ➔ Very **quick** to **search for the details of a book** and match to who has borrowed it.
- Accuracy** - ➔ Databases can be set up to **warn you when books become overdue**. This means that the Librarian can contact the borrower to request the book's return.
- Space** - ➔ One computer database can store the **details of many thousands of books**. This means that all of the old filing drawers can be removed **leaving physical room for more books**.
- Cheaper** - ➔ **One database** can handle the **same amount of work as many librarians**. This saves on employment costs.



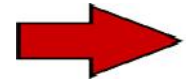
Old style library cards were **easy to mix up** and could be **difficult to find**



Library databases can **quickly and easily locate book records** and **find out who has borrowed them**



Applications



Links to Theory Units:

[Section 1: Types and Components of Computer Systems](#)

Section 4: Computer Networks

[Section 7: The Ways in which ICT is used](#)

[Section 2: Input and Output Devices](#)

[Section 5: Data Types](#)

[Section 7: Part 2](#)

[Section 3: Storage Devices and Media](#)

[Section 6: The Effects of Using ICT](#)

[Section 8: Systems Analysis and Design](#)

Links to Practical Units:

[Section 9: Communication](#)

Section 12: Integration

[Section 15: Web site Authoring](#)

[Section 10: Document Production](#)

Section 13: Output Data

[Section 16: Presentation Authoring](#)

[Section 11: Data Manipulation](#)

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